

Spice Avenue

Developer Responsibility Document

Chapter 1 – Project Development Structure

1. Project Overview

Project Name

Spice Avenue – Web-Based Food Ordering System

Project Objective

To develop a centralized web-based food ordering system for a restaurant company with multiple branches that enables customers to place pickup and delivery orders while allowing restaurant staff to efficiently manage menus, branches, orders, deliveries, complaints, and customer reviews.

2. Development Philosophy

Instead of dividing the project randomly, the system will be divided according to **business functions**.

Each member is responsible for **one complete business module**, including:

- Database implementation (their owned tables and relationships)
- Backend development
- Frontend development
- CRUD operations
- Validation
- Business rules
- Integration with other modules
- Testing

Every member will complete a full-featured module that can be individually demonstrated during the viva.

3. Shared Responsibilities (Everyone)

These activities are completed together before individual development begins.

Requirement Analysis

- Finalize system scope
- Finalize business rules
- Finalize workflows
- Approve functional requirements

Database Design

The entire team designs the database together.

This includes:

- ER Diagram
- Tables
- Relationships
- Primary Keys
- Foreign Keys
- Naming conventions

After the design is finalized:

One shared SQL script is created.

Every member imports the same database.

No member should create a separate database.

UI Design

The team agrees on:

- Color palette
- Navigation layout

- Typography
- Button styles
- Form styles
- Table styles
- Dashboard layout

This keeps the system consistent.

Project Setup

The team creates:

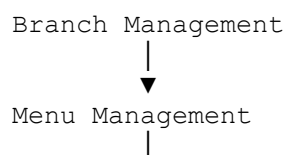
- GitHub Repository
 - Spring Boot Project
 - MySQL Database
 - Shared dependencies
 - Folder structure
-

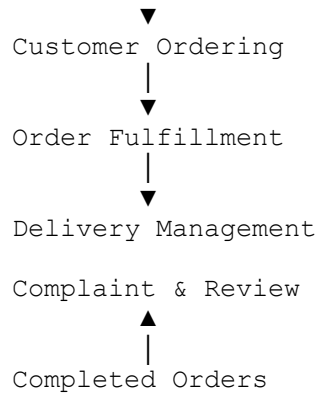
4. Module Allocation

Member	Module
Member 1	Branch Management
Member 2	Menu Management
Member 3	Customer Ordering
Member 4	Order Fulfillment Management
Member 5	Delivery Management
Member 6	Complaint & Review Management

Each member owns **one business process**.

5. Module Dependencies





This means:

Branch Management supports Menu Management.

Menu Management supports Customer Ordering.

Customer Ordering produces Orders.

Order Fulfillment processes Orders.

Delivery completes Orders.

Complaint & Review begins only after completed orders.

6. Supporting Features

The following are **shared system functionalities** and are **not assigned as individual modules**:

- Authentication
- User Account Management
- Role-Based Access Control
- Notifications
- Card Payment Integration
- Password Management

All members will use these features where necessary.

7. Development Standards

Every member should implement:

Frontend

- Responsive pages
 - Form validation
 - Search
 - Tables
 - Confirmation dialogs
 - Success/error messages
-

Backend

- REST APIs
 - Service layer
 - Repository layer
 - DTOs
 - Validation
 - Exception handling
-

Database

Every module should:

- Use proper foreign keys
 - Follow normalization
 - Avoid duplicated data
 - Maintain referential integrity
-

8. GitHub Workflow

main

↓

develop

↓

member-1

member-2

member-3
member-4
member-5
member-6

Each member works on their own branch.

After testing:

Merge into **develop**

Finally merge into **main**

9. Integration Strategy

Development occurs independently.

Testing occurs together.

The order of integration is:

1. Branch Management
2. Menu Management
3. Customer Ordering
4. Order Fulfillment
5. Delivery Management
6. Complaint & Review

This order follows the business workflow and reduces integration conflicts.

10. Individual Responsibility Principle

Each member is responsible for:

- Designing their module
- Building the frontend
- Developing the backend
- Implementing CRUD operations
- Applying business rules
- Testing their module
- Fixing defects

- Demonstrating their work during the viva

However, **all members are collectively responsible for the final integrated system.**

CHAPTER 2

MEMBER 1 – BRANCH MANAGEMENT

Module Information

Item	Description
Module Name	Branch Management
Primary User	Operations Manager, Branch Manager
Priority	High
Module Type	Core Business Function
Depends On	Authentication
Used By	Menu Management, Customer Ordering, Order Fulfillment, Delivery Management

1. Module Purpose

The Branch Management module is responsible for managing all restaurant branches and their operational settings. It ensures that each branch has accurate information, assigned management, operating hours, and delivery areas so that customers can order only from eligible branches.

This module serves as the foundation of the entire system because every menu item, customer order, and delivery belongs to a specific branch.

2. Main Responsibilities

This module is responsible for:

- Managing restaurant branches
- Managing branch information
- Managing branch operating hours
- Managing delivery areas
- Assigning Branch Managers
- Monitoring branch performance

- Activating and deactivating branches

3. NOT Responsible For

This module **does NOT**:

- ✗ Create menu items
- ✗ Manage categories
- ✗ Accept customer orders
- ✗ Prepare food
- ✗ Deliver food
- ✗ Resolve complaints

Those belong to other modules.

4. Sub Functions

4.1 Create Branch

Purpose

Allows the Operations Manager to register a new restaurant branch.

Frontend

Page

Add Branch

Fields

Branch Name

Street Address

Contact Number

Email

Opening Time

Closing Time

Status

Buttons

Save

Cancel

Backend

Create Branch API

Validation

Save Branch

Return Success

Database

Table

Branch

Insert

Branch Name

Address

Phone

Email

Opening Time

Closing Time

Status

CRUD

Create ✓

Read ✕

Update ✕

Delete ✕

Business Rules

Branch name cannot be empty.

Phone number must be unique.

Email must be unique.

Branch starts as Active.

4.2 View Branch Details

Purpose

View all branch information.

Frontend

Branch List

Search

View Details

Can View

Branch Name

Address

Contact

Operating Hours

Assigned Manager

Delivery Areas

Status

Performance Summary

CRUD

Read ✓

Business Rules

Everyone with permission can view.

Only Operations Manager can view every branch.

Branch Manager can only view their assigned branch.

4.3 Update Branch Information

Purpose

Edit existing branch details.

Frontend

Edit Branch

Editable

Phone

Email

Address

Hours

Status

CRUD

Update ✓

Business Rules

Branch ID cannot change.

Branch Name should remain unique.

4.4 Activate / Deactivate Branch

Purpose

Temporarily disable a branch.

CRUD

Update

(Not Delete)

Reason

Historical data should remain.

Orders should never disappear.

Business Rule

Inactive branches

Cannot receive new orders.

Still appear in reports.

4.5 Assign Branch Manager

Purpose

Assign one Branch Manager to one branch.

Frontend

Dropdown

Available Managers

CRUD

Update

Business Rules

One manager

↓

One branch

A manager cannot manage multiple branches simultaneously.

Database

Branch

Manager_ID

4.6 Manage Operating Hours

Purpose

Change

Opening

Closing

Times

CRUD

Update

Example

Monday

8 AM

10 PM

Business Rule

Closing time must be later than opening time.

4.7 Manage Delivery Areas

This is one of our custom features.

Purpose

Allow each branch to define where it delivers.

Instead of calculating GPS distance,
the Branch Manager manages a list of supported delivery areas.

Example

Negombo Town

Kurana

Katunayake

Seeduwa

Ja-Ela

Customers simply select one of these areas when placing a delivery order.

Frontend

Delivery Areas

Area List

Add Area

Edit Area

Deactivate Area

Fields

Area Name

Status

Backend

CRUD APIs

Validation

Database

Table

DeliveryArea

Columns

Area_ID

Branch_ID

Area_Name

Status

CRUD

Create ✓

Read ✓

Update ✓

Delete (Soft Delete)

Business Rules

Area belongs to ONE branch.

Different branches can have different delivery areas.

Inactive areas cannot be selected during ordering.

4.8 View Branch Performance

Purpose

Provide operational statistics.

Information

Total Orders

Total Revenue

Completed Orders

Cancelled Orders

Pickup Orders

Delivery Orders

Average Rating

Complaint Count

Most Popular Menu Items

Frontend

Dashboard

Charts

Tables

Summary Cards

Backend

Read Only

CRUD

Read

Business Rules

Cannot modify report.

Only Operations Manager.

Branch Manager sees only their own branch.

5. Database Tables Used

Primary

Branch

DeliveryArea

Reference

User

Order

Review

Complaint

6. APIs

POST /branches

GET /branches

GET /branches/{id}

PUT /branches/{id}

PATCH /branches/status

PATCH /branches/hours

POST /delivery-areas

GET /delivery-areas

PUT /delivery-areas

DELETE /delivery-areas

GET /branch-performance

7. Frontend Pages

Branch List

Add Branch

Edit Branch

Branch Details

Assign Manager

Delivery Area Management

Operating Hours

Performance Dashboard

8. Integration

This module provides data to:

☒ Menu Management

(Menu belongs to Branch)

☒ Customer Ordering

(Customer selects Branch)

✓ Order Fulfillment

(Orders belong to Branch)

✓ Delivery Management

(Riders deliver from Branch)

9. Testing Checklist

Test creating a branch.

Test duplicate email.

Test duplicate phone.

Test invalid hours.

Test assigning a manager.

Test adding delivery areas.

Test inactive branch.

Test inactive delivery area.

Test reports.

10. Deliverables

At the end of this module, Member 1 should have completed:

- Branch CRUD
- Delivery Area CRUD
- Operating Hours Management
- Branch Activation/Deactivation

- Branch Manager Assignment
- Branch Performance Dashboard
- Frontend screens
- Backend APIs
- Database integration
- Validation
- Testing

CHAPTER 3

MEMBER 2 – MENU MANAGEMENT

Module Information

Item	Description
Module Name	Menu Management
Primary User	Branch Manager
Priority	High
Module Type	Core Business Function
Depends On	Branch Management
Used By	Customer Ordering, Order Fulfillment

1. Module Purpose

The Menu Management module enables the **Branch Manager** to create and maintain the menu of their assigned branch. It ensures that customers always view accurate menu information, including food categories, menu items, pricing, availability, images, and optional item variations.

Each branch manages **its own menu**, allowing different branches to offer different food items, prices, and availability.

2. Main Responsibilities

This module is responsible for:

- Managing Food Categories
- Managing Menu Items
- Managing Item Variations
- Managing Menu Pricing
- Managing Menu Availability
- Uploading Food Images
- Searching Menu Items

3. NOT Responsible For

This module **does NOT**:

- ✗ Manage restaurant branches
- ✗ Process customer orders
- ✗ Manage shopping carts
- ✗ Assign delivery riders
- ✗ Prepare food
- ✗ Deliver orders
- ✗ Handle complaints

These belong to other modules.

4. Sub Functions

4.1 Manage Food Categories

Purpose

Allows the Branch Manager to organize menu items into categories.

Examples:

Pizza

Burgers

Rice

Beverages

Desserts

Frontend

Page

Category Management

Fields

Category Name

Description

Status

Buttons

Add

Update

Deactivate

Search

Backend

- Category CRUD APIs
 - Validation
 - Duplicate checking
-

Database

Table

Category

Columns

Category_ID

Branch_ID

Category_Name

Description

Status

CRUD

Create

Create new category.

Read

View category list.

Search categories.

Update

Edit category details.

Delete

Soft delete (Deactivate).

Business Rules

- Category names must be unique **within the same branch**.
 - Cannot permanently delete a category.
 - Cannot deactivate a category that still contains active menu items.
-

4.2 Manage Menu Items

Purpose

Allows the Branch Manager to create and maintain food items.

Frontend

Page

Menu Item Management

Fields

Food Name

Category

Description

Base Price

Food Image

Status

Buttons

Add

Update

Deactivate

Search

Backend

Menu CRUD

Validation

Search APIs

Database

Table

Menu_Item

Columns

Menu_ID

Category_ID

Branch_ID

Food_Name

Description

Image

Status

CRUD

Create

Create menu item.

Read

View menu items.

Search.

Filter.

Update

Update

Name

Description

Category

Image

Delete

Deactivate menu item.

Business Rules

- Food name cannot be empty.
 - Food belongs to one category.
 - Food belongs to one branch.
 - Inactive menu items cannot be ordered.
-

4.3 Manage Item Variations

(This is one of our finalized features.)

Purpose

Allows menu items to have optional variations.

Example:

Chicken Pizza

Small

Medium

Large

or

Soft Drink

500ml

1L

1.5L

Not every menu item requires variations.

Example:

French Fries

(No variations)

This feature is optional per menu item.

Frontend

Variation Management

Fields

Variation Name

Additional Price

Status

Buttons

Add

Update

Remove

Backend

Variation CRUD

Validation

Database

Table

Menu_Variation

Columns

Variation_ID

Menu_ID

Variation_Name

Additional_Price

Status

CRUD

Create

Create variation.

Read

View variations.

Update

Update variation.

Delete

Delete variation.

Business Rules

- Variations belong to one menu item.
 - Menu items may have zero, one, or many variations.
 - Additional price cannot be negative.
-

4.4 Manage Pricing

Purpose

Allows the Branch Manager to define menu prices.

If a menu item has variations, each variation can have its own price adjustment.

CRUD

Update

(Read pricing information)

Business Rules

- Prices cannot be negative.
 - Price changes immediately affect new customer orders.
 - Existing orders retain their original prices.
-

4.5 Manage Availability

Purpose

Allows the Branch Manager to control whether a menu item is available for ordering.

Available

↓

Out of Stock

↓

Available

Frontend

Dropdown

Available

Out of Stock

CRUD

Update

Business Rules

- Out of Stock items remain visible but cannot be ordered.
 - Customers must clearly see the availability status.
-

4.6 Upload Food Images

Purpose

Allows the Branch Manager to upload or replace menu item images.

CRUD

Create

Read

Update

Delete (Replace/Remove)

Business Rules

- One primary image per menu item.
 - Supported image formats only (e.g., JPG, PNG).
-

4.7 Search Menu Items

Purpose

Quickly locate menu items.

Search By

- Name
 - Category
 - Availability
-

CRUD

Read

5. Database Tables Used

Primary Tables

Category

Menu_Item

Menu_Variation

Reference Tables

Branch

DeliveryArea

(Menu items belong to a specific branch.)

6. APIs

POST /categories

GET /categories

PUT /categories/{id}

PATCH /categories/status

POST /menu-items

GET /menu-items

GET /menu-items/{id}

PUT /menu-items/{id}

PATCH /menu-items/status

POST /variations

GET /variations

PUT /variations/{id}

DELETE /variations/{id}

GET /menu-items/search

7. Frontend Pages

Category Management

Menu Item Management

Variation Management

Pricing Management

Availability Management

Image Upload

Search Menu

8. Integration

Provides data to:

Customer Ordering

- Display menu
 - Search food
 - Show prices
 - Show images
 - Show variations
 - Check availability
-

Order Fulfillment

- View ordered menu items
 - Confirm item details
 - Check availability if needed
-

Depends on:

Branch Management

- Every menu item belongs to one branch.
 - Every category belongs to one branch.
-

9. Testing Checklist

- ☒ Create category
- ☒ Duplicate category validation
- ☒ Create menu item
- ☒ Update menu item
- ☒ Upload image

- ✓ Add variation
 - ✓ Delete variation
 - ✓ Mark item Out of Stock
 - ✓ Search menu
 - ✓ Verify customer cannot order Out of Stock item
-

10. Deliverables

At the end of this module, Member 2 should have completed:

- Food Category CRUD
 - Menu Item CRUD
 - Item Variation CRUD
 - Pricing Management
 - Availability Management
 - Food Image Upload
 - Menu Search
 - Frontend pages
 - Backend APIs
 - Database integration
 - Validation
 - Testing
-



Difference Between Member 1 and Member 2

This is something you can explain during your viva if asked.

Branch Management (Member 1) Menu Management (Member 2)

Manages restaurant branches

Manages food menu

Creates branches

Creates menu items

Manages delivery areas

Manages food categories

Assigns branch managers

Sets food prices

Branch Management (Member 1) Menu Management (Member 2)

Sets operating hours

Manages item variations

Views branch performance

Controls food availability

Used before menus exist

Depends on an existing branch

In simple terms:

- **Member 1 answers:** *"Where does the restaurant operate?"*
- **Member 2 answers:** *"What food does each branch sell?"*

These are different business domains, so there is no overlap.

CHAPTER 4

MEMBER 3 – CUSTOMER ORDERING

Module Information

Item	Description
Module Name	Customer Ordering
Primary User	Customer
Priority	Very High
Module Type	Core Business Function
Depends On	Branch Management, Menu Management
Used By	Order Fulfillment Management

1. Module Purpose

The Customer Ordering module allows customers to browse restaurant menus, place pickup or delivery orders, manage delivery addresses, select menu item variations, manage their shopping cart, and monitor their orders.

This module represents the complete ordering experience from selecting an order type until the order is successfully submitted.

2. Main Responsibilities

This module is responsible for:

- Selecting the order type
- Managing delivery addresses
- Selecting delivery areas
- Selecting branches
- Browsing menus
- Searching food
- Viewing food details
- Selecting item variations

- Managing the shopping cart
 - Checkout
 - Placing orders
 - Cancelling eligible orders
 - Tracking order status
 - Viewing order history
-

3. NOT Responsible For

This module **does NOT**:

- ✗ Confirm orders
- ✗ Prepare food
- ✗ Assign delivery riders
- ✗ Update order statuses
- ✗ Deliver food
- ✗ Resolve complaints

Those belong to later modules.

4. Sub Functions

4.1 Select Order Type

Purpose

The customer chooses how they wish to receive the food.

Options

Pickup

Delivery

Frontend

First screen after browsing starts.

Buttons

Pickup

Delivery

Backend

Store selected order type in the current order.

Database

Stored in

Order

Order_Type

CRUD

Update

(The customer selects one option.)

Business Rules

- Exactly one order type must be selected.
 - Once items are added to the cart, changing the order type clears the cart.
-

4.2 Manage Delivery Addresses

Purpose

Customers manage their saved delivery addresses.

Example

Home

Office

Boarding House

Frontend

Page

My Delivery Addresses

Fields

Address Label

House Number

Street

Delivery Area

Buttons

Add

Edit

Delete

Backend

Address CRUD APIs

Database

Table

Customer_Address

Columns

Address_ID

Customer_ID

Label

House_Number

Street

DeliveryArea_ID

CRUD

Create

Save a new delivery address.

Read

View saved addresses.

Update

Modify an address.

Delete

Remove saved address.

Business Rules

- Customer may save multiple addresses.
 - One address can be marked as default.
 - Delivery Area must exist in the selected branch's supported areas.
-

4.3 Use Temporary Address

Purpose

Allows customers to use a one-time delivery address without saving it.

Frontend

Checkbox

Use Temporary Address

Fields

House Number

Street

Delivery Area

CRUD

Create (Temporary)

Business Rules

- Temporary addresses are **not stored** in the customer's profile.
 - Used only for the current order.
-

4.4 Select Delivery Area

Purpose

The customer selects one of the delivery areas supported by the chosen branch.

Frontend

Dropdown

Negombo Town

Kurana

Katunayake

Seeduwa

Ja-Ela

CRUD

Read

Business Rules

- Only active delivery areas are displayed.
 - If no delivery areas are available, delivery ordering is not allowed for that branch.
-

4.5 Select Branch

Purpose

Allows customers to choose the branch that will prepare the order.

Frontend

Branch List

Displays

Branch Name

Address

Operating Hours

CRUD

Read

Business Rules

Pickup

→ Customer may choose any active branch.

Delivery

→ Only branches serving the selected delivery area are displayed.

4.6 Browse Menu

Purpose

Display menu items for the selected branch.

Displays

- Categories
- Menu Items
- Images

- Prices
 - Availability
-

CRUD

Read

Business Rules

- Customers cannot browse menus until a branch has been selected.
 - Out of Stock items remain visible but cannot be added to the cart.
-

4.7 Search Food Items

Purpose

Allows customers to quickly find food items.

Search By

- Food Name
 - Category
-

CRUD

Read

4.8 View Food Details

Purpose

Display detailed information about a menu item.

Shows

Food Name

Description

Image

Price

Availability

Variations

CRUD

Read

4.9 Select Item Variation

Purpose

Allows customers to choose an available variation before adding the item to the cart.

Example

Chicken Pizza

○ Small

○ Medium

○ Large

CRUD

Update

Business Rules

- If a menu item has variations, one must be selected.
 - If no variations exist, this step is skipped.
-

4.10 Manage Shopping Cart

Purpose

Customers manage the items they intend to purchase.

Frontend

Shopping Cart

Displays

- Items
 - Variations
 - Quantity
 - Price
 - Total
-

Buttons

Add Item

Update Quantity

Remove Item

Clear Cart

Database

Table

Shopping_Cart

Table

Cart_Item

CRUD

Create

Add item

Read

View cart

Update

Update quantity

Delete

Remove item

Clear cart

Business Rules

- Quantity must be greater than zero.
 - Cannot add Out of Stock items.
 - Total updates automatically.
-

4.11 Checkout

Purpose

Review all order details before submitting the order.

Displays

- Branch
 - Order Type
 - Delivery Address (if applicable)
 - Cart Items
 - Total Price
 - Selected Payment Method
-

Buttons

Place Order

Cancel

CRUD

Read

Business Rules

- Cart cannot be empty.
 - Customer must review all details before placing the order.
-

4.12 Place Order

Purpose

Submit the order to the selected branch.

Backend

Creates

Order

Order Items

Database

Tables

Order

Order_Item

CRUD

Create

Business Rules

- Initial order status is **Pending**.
 - Estimated preparation time will be set later by the Branch Manager.
-

4.13 Cancel Order

Purpose

Allow customers to cancel eligible orders.

CRUD

Update

Business Rules

Order can only be cancelled when status is:

Pending

Confirmed

Once the order status changes to:

Preparing

Cancellation is no longer allowed.

4.14 Track Order Status

Purpose

Allow customers to monitor order progress.

Possible Statuses

Pickup

Pending

Confirmed

Preparing

Ready for Pickup

Picked Up

Delivery

Pending

Confirmed

Preparing

Ready for Delivery

Out for Delivery

Delivered

CRUD

Read

4.15 View Order History

Purpose

Display all previous customer orders.

Shows

- Order Number
 - Date
 - Branch
 - Total
 - Status
-

CRUD

Read

Business Rules

Only the logged-in customer can view their own order history.

5. Database Tables Used

Primary Tables

Customer_Address

Shopping_Cart

Cart_Item

Order

Order_Item

Reference Tables

Branch

DeliveryArea

Menu_Item

Menu_Variation

Category

6. APIs

POST /addresses
GET /addresses
PUT /addresses/{id}
DELETE /addresses/{id}

GET /branches
GET /delivery-areas

GET /menu
GET /menu/search

POST /cart/items
GET /cart
PUT /cart/items/{id}
DELETE /cart/items/{id}
DELETE /cart

POST /orders
GET /orders
GET /orders/history
PATCH /orders/{id}/cancel

7. Frontend Pages

Order Type Selection

Delivery Address Management

Branch Selection

Menu Page

Food Details

Shopping Cart

Checkout

Order Confirmation

Track Order

Order History

8. Integration

Provides data to:

Order Fulfillment Management

- Newly placed orders
 - Customer details
 - Selected menu items
 - Delivery address
 - Order type
 - Payment method
-

Depends on:

Branch Management

- Active branches
- Delivery areas

Menu Management

- Categories
 - Menu items
 - Variations
 - Pricing
 - Availability
-

9. Testing Checklist

- ✓ Select Pickup and Delivery
 - ✓ Add delivery address
 - ✓ Use temporary address
 - ✓ Select delivery area
 - ✓ Select branch
 - ✓ Browse menu
 - ✓ Search food
 - ✓ Select item variation
 - ✓ Add items to cart
 - ✓ Update quantity
 - ✓ Remove item
 - ✓ Clear cart
 - ✓ Place order
 - ✓ Cancel eligible order
 - ✓ Track order
 - ✓ View order history
-

10. Deliverables

At the end of this module, Member 3 should have completed:

- Delivery Address CRUD
- Shopping Cart CRUD
- Branch Selection
- Delivery Area Selection
- Menu Browsing
- Menu Search
- Item Variation Selection
- Checkout
- Order Placement
- Order Cancellation
- Order Tracking
- Order History
- Frontend pages
- Backend APIs
- Database integration
- Validation
- Testing



Difference Between Member 3 and Member 4

This is another question lecturers love to ask.

Member 3 – Customer Ordering Member 4 – Order Fulfillment

Customer places the order	Branch Manager receives the order
Manages cart	Confirms or rejects orders
Selects food	Prepares food
Selects branch	Updates order status
Chooses address	Sets preparation time
Places order	Assigns delivery rider
Tracks order	Marks Ready for Pickup/Delivery

CHAPTER 5

MEMBER 4 – ORDER FULFILLMENT MANAGEMENT

Module Information

Item	Description
Module Name	Order Fulfillment Management
Primary User	Branch Manager
Priority	Very High
Module Type	Core Business Function
Depends On	Customer Ordering
Used By	Delivery Management

1. Module Purpose

The Order Fulfillment Management module enables the **Branch Manager** to receive customer orders, review them, confirm or reject them, manage the food preparation process, estimate preparation time, assign delivery riders, and prepare orders for pickup or delivery.

This module manages the restaurant's internal workflow after an order has been placed by the customer.

2. Main Responsibilities

This module is responsible for:

- Viewing incoming orders
- Confirming orders
- Rejecting orders
- Updating order statuses
- Managing food preparation
- Setting estimated preparation time

- Preparing orders for pickup
 - Preparing orders for delivery
 - Assigning delivery riders
 - Cancelling orders when necessary
-

3. NOT Responsible For

This module **does NOT**:

✗ Create customer orders

✗ Manage menus

✗ Manage branches

✗ Deliver food

✗ Handle complaints

Those belong to other modules.

4. Sub Functions

4.1 View Incoming Orders

Purpose

Allows the Branch Manager to view all newly received customer orders for their assigned branch.

Frontend

Page

Incoming Orders

Displays

Order Number

Customer Name

Order Type

Order Date

Total Amount

Current Status

Buttons

View Details

Confirm

Reject

Backend

Retrieve all **Pending** orders assigned to the branch.

Database

Table

Order

CRUD

Read

View incoming orders.

Business Rules

- Branch Managers can only view orders belonging to their assigned branch.
 - Orders are displayed with the newest first.
-

4.2 Confirm Order

Purpose

Accept the customer's order and begin processing it.

Frontend

Button

Confirm Order

Backend

Update order status

Pending

↓

Confirmed

CRUD

Update

Business Rules

- Only Pending orders can be confirmed.
 - Confirmation triggers customer notification.
-

4.3 Reject Order

Purpose

Reject customer orders when they cannot be fulfilled.

Examples

- Branch closed
 - Required ingredients unavailable
 - Operational issue
-

Frontend

Button

Reject Order

Field

Reason

CRUD

Update

Business Rules

- Only Pending orders can be rejected.
 - A rejection reason is mandatory.
 - Customer receives a notification.
-

4.4 Update Order Status

Purpose

Manage the progress of an order during preparation.

Available Statuses

Pickup

Confirmed

↓

Preparing

↓

Ready for Pickup

↓

Picked Up

Delivery

Confirmed

↓

Preparing

↓

Ready for Delivery

CRUD

Update

Business Rules

Status progression must follow the correct workflow.

Statuses cannot be skipped.

4.5 Set Estimated Preparation Time

Purpose

Provide customers with an estimated preparation time after confirming the order.

Frontend

Field

Estimated Preparation Time
(minutes)

Example

20 Minutes

CRUD

Update

Business Rules

- Must be greater than 0 minutes.
 - Can only be entered once the order is confirmed.
 - Customers can view the estimate.
-

4.6 Mark as Preparing

Purpose

Indicate that food preparation has started.

Status

Confirmed

↓

Preparing

CRUD

Update

Business Rules

Once an order becomes **Preparing**, the customer can no longer cancel it.

4.7 Mark Ready for Pickup

Purpose

Indicate that a pickup order is ready for customer collection.

Status

Preparing

↓

Ready for Pickup

CRUD

Update

Business Rules

Only available for Pickup orders.

4.8 Mark Picked Up

Purpose

Confirm that the customer has collected the pickup order.

Status

Ready for Pickup

↓

Picked Up

CRUD

Update

Business Rules

- Only the Branch Manager can mark an order as Picked Up.
 - This completes the pickup workflow.
-

4.9 Mark Ready for Delivery

Purpose

Notify that a delivery order is packed and ready to be handed over to a Delivery Rider.

Status

Preparing

↓

Ready for Delivery

CRUD

Update

Business Rules

Only available for Delivery orders.

4.10 Assign Delivery Rider

Purpose

Assign an available Delivery Rider to a delivery order.

Frontend

Dropdown

Available Riders

Displays

Rider Name

Availability

Button

Assign Rider

Backend

Retrieve riders whose status is

Available

Assign selected rider to the order.

Database

Tables

Order

Delivery

User

CRUD

Update

Business Rules

- Only riders with **Available** status can be assigned.
 - One rider per delivery order.
 - Rider status automatically changes to **Busy** after assignment.
-

4.11 Cancel Order

Purpose

Allow the Branch Manager to cancel orders when necessary.

Examples

- Equipment failure
- Ingredient shortage
- Operational emergency

CRUD

Update

Business Rules

Cancelled orders remain stored for reporting purposes.

Customer receives a cancellation notification.

5. Database Tables Used

Primary Tables

Order

Order_Item

Delivery

Reference Tables

Branch

Customer

User

Delivery_Rider

6. APIs

GET /orders/incoming

GET /orders/{id}

PATCH /orders/{id}/confirm

PATCH /orders/{id}/reject

PATCH /orders/{id}/status

PATCH /orders/{id}/preparing

PATCH /orders/{id}/ready-pickup

PATCH /orders/{id}/picked-up

PATCH /orders/{id}/ready-delivery

PATCH /orders/{id}/assign-rider

PATCH /orders/{id}/cancel

7. Frontend Pages

Incoming Orders

Order Details

Preparation Dashboard

Estimated Preparation Time

Assign Delivery Rider

Pickup Queue

Delivery Queue

8. Integration

Receives Data From

Customer Ordering

- New Orders

- Order Items
 - Customer Details
 - Delivery Address
 - Order Type
-

Sends Data To

Delivery Management

- Assigned Rider
 - Delivery Order
 - Ready for Delivery Status
-

Uses Data From

Menu Management

- Ordered Menu Items
 - Item Variations
 - Prices
-

Uses Data From

Branch Management

- Branch Information
 - Branch Manager Assignment
-

9. Testing Checklist

- ☒ View incoming orders
- ☒ Confirm order
- ☒ Reject order with reason

- ✓ Prevent rejection without reason
 - ✓ Set preparation time
 - ✓ Update Preparing status
 - ✓ Mark Ready for Pickup
 - ✓ Mark Picked Up
 - ✓ Mark Ready for Delivery
 - ✓ Assign available rider
 - ✓ Prevent assigning busy rider
 - ✓ Cancel order
-

10. Deliverables

At the end of this module, Member 4 should have completed:

- Incoming Order Management
 - Order Confirmation
 - Order Rejection
 - Order Status Management
 - Estimated Preparation Time
 - Pickup Workflow Management
 - Delivery Preparation Workflow
 - Delivery Rider Assignment
 - Order Cancellation
 - Frontend pages
 - Backend APIs
 - Database integration
 - Validation
 - Testing
-

Difference Between Member 4 and Member 5

This is another common viva question.

Member 4 – Order Fulfillment Member 5 – Delivery Management

Receives customer orders	Receives assigned delivery orders
Confirms or rejects orders	Accepts delivery assignments
Manages food preparation	Delivers food
Sets preparation time	Updates delivery progress
Marks Ready for Pickup	Marks Out for Delivery
Marks Ready for Delivery	Marks Delivered
Assigns delivery riders	Manages rider availability

Simple explanation

- **Member 4 answers:** *"What happens inside the restaurant after an order is placed?"*
- **Member 5 answers:** *"What happens after the food leaves the restaurant?"*

This creates a clean separation between restaurant operations and delivery operations, ensuring each member owns a complete business module without overlapping responsibilities.

CHAPTER 6

MEMBER 5 – DELIVERY MANAGEMENT

Module Information

Item	Description
Module Name	Delivery Management
Primary User	Delivery Rider
Priority	High
Module Type	Core Business Function
Depends On	Order Fulfillment Management
Used By	Complaint & Review Management

1. Module Purpose

The Delivery Management module enables **Delivery Riders** to manage assigned delivery orders from the moment they are assigned by the Branch Manager until the order is successfully delivered to the customer. It allows riders to manage their availability, accept delivery assignments, update delivery progress, and maintain their delivery history.

2. Main Responsibilities

This module is responsible for:

- Managing rider availability
 - Viewing assigned deliveries
 - Accepting delivery assignments
 - Updating delivery status
 - Marking orders as Out for Delivery
 - Marking orders as Delivered
 - Viewing delivery history
-

3. NOT Responsible For

This module **does NOT**:

- ✗ Receive customer orders
- ✗ Confirm or reject orders
- ✗ Prepare food
- ✗ Assign delivery riders
- ✗ Manage menu items
- ✗ Handle complaints

These belong to other modules.

4. Sub Functions

4.1 Manage Rider Availability

Purpose

Allows Delivery Riders to indicate whether they are available to receive delivery assignments.

Frontend

Page

My Availability

Options

Available

Busy

Offline

Buttons

Update Status

Backend

Update the rider's availability status.

Database

Table

User

Column

Availability_Status

CRUD

Update

Update rider availability.

Read

View current availability.

Business Rules

- Riders marked **Available** can receive delivery assignments.
- Riders marked **Busy** cannot receive new assignments.
- Riders marked **Offline** are hidden from assignment lists.
- Rider status automatically changes to **Busy** after accepting a delivery.
- Rider status automatically changes back to **Available** after completing a delivery.

4.2 View Assigned Deliveries

Purpose

Allows Delivery Riders to view delivery orders assigned to them.

Frontend

Page

Assigned Deliveries

Displays

Order Number

Customer Name

Branch

Delivery Address

Delivery Area

Order Status

Buttons

View Details

Accept Delivery

Backend

Retrieve all assigned delivery orders for the logged-in rider.

Database

Tables

Delivery

Order

CRUD

Read

View assigned deliveries.

Business Rules

- Riders can only view deliveries assigned to them.
 - Only delivery orders are displayed.
-

4.3 Accept Delivery Assignment

Purpose

Allows a Delivery Rider to accept a delivery assigned by the Branch Manager.

Frontend

Button

Accept Delivery

Backend

Update assignment status.

CRUD

Update

Accept assignment.

Business Rules

- Only assigned riders can accept the delivery.
 - Once accepted, the rider's availability changes to **Busy**.
-

4.4 Update Delivery Status

Purpose

Allows the Delivery Rider to update the progress of the delivery.

Available Statuses

Ready for Delivery

↓

Out for Delivery

↓

Delivered

CRUD

Update

Update delivery progress.

Business Rules

- Status progression must follow the correct sequence.
 - Statuses cannot be skipped.
-

4.5 Mark Out for Delivery

Purpose

Indicates that the rider has collected the order from the branch and is travelling to the customer's location.

Frontend

Button

Mark Out for Delivery

Backend

Update delivery status.

CRUD

Update

Update status.

Business Rules

- Only orders marked **Ready for Delivery** can be changed to **Out for Delivery**.
 - Customer receives an in-system notification.
-

4.6 Mark Order as Delivered

Purpose

Confirms that the customer has successfully received the order.

Frontend

Button

Mark Delivered

Backend

Update delivery status to **Delivered**.

Automatically update rider availability.

CRUD

Update

Update delivery status.

Business Rules

- Only orders marked **Out for Delivery** can be marked **Delivered**.
 - Customer receives a delivery confirmation notification.
 - Rider availability automatically changes back to **Available**.
 - Delivered orders become eligible for customer reviews and complaints.
-

4.7 View Delivery History

Purpose

Allows Delivery Riders to review previously completed deliveries.

Frontend

Page

Delivery History

Displays

Order Number

Customer

Branch

Delivery Date

Status

Backend

Retrieve completed deliveries.

CRUD

Read

View delivery history.

Business Rules

- Riders can only view their own delivery history.
 - Completed deliveries cannot be edited.
-

5. Database Tables Used

Primary Tables

Delivery

Reference Tables

Order

User

Branch

6. APIs

GET /deliveries/assigned

GET /deliveries/history

PATCH /rider/availability

PATCH /deliveries/{id}/accept

PATCH /deliveries/{id}/out-for-delivery

PATCH /deliveries/{id}/delivered

7. Frontend Pages

My Availability

Assigned Deliveries

Delivery Details

Delivery History

8. Integration

Receives Data From

Order Fulfillment Management

- Assigned delivery orders
 - Customer delivery address
 - Assigned branch
 - Rider assignment
-

Sends Data To

Complaint & Review Management

- Completed delivery status
 - Delivery completion time
-

Uses Data From

Customer Ordering

- Delivery address
 - Delivery area
-

Uses Data From

Branch Management

- Branch details
-

9. Testing Checklist

- ✓ Update rider availability
- ✓ Prevent assigning Offline riders
- ✓ View assigned deliveries
- ✓ Accept delivery assignment

- ✓ Mark Out for Delivery
 - ✓ Prevent invalid status changes
 - ✓ Mark Delivered
 - ✓ Automatic availability update after delivery
 - ✓ View delivery history
-

10. Deliverables

At the end of this module, Member 5 should have completed:

- Rider Availability Management
 - Assigned Delivery Management
 - Delivery Assignment Acceptance
 - Delivery Status Management
 - Delivery Completion
 - Delivery History
 - Frontend pages
 - Backend APIs
 - Database integration
 - Validation
 - Testing
-

Difference Between Member 5 and Member 6

Another common viva question.

Member 5 – Delivery Management Member 6 – Complaint & Review Management

Delivers customer orders

Handles feedback after completion

Manages rider availability

Manages complaints

Updates delivery status

Resolves complaints

Marks orders as Delivered

Manages reviews and ratings

Delivery ends here

Customer support begins here

Simple explanation

- **Member 5 answers:** *"How does the restaurant deliver food to the customer?"*
- **Member 6 answers:** *"What happens after the customer receives the food?"*

CHAPTER 7

MEMBER 6 – COMPLAINT & REVIEW MANAGEMENT

Module Information

Item	Description
Module Name	Complaint & Review Management
Primary User	Customer, Customer Service Supervisor
Priority	High
Module Type	Core Business Function
Depends On	Delivery Management
Used By	Branch Management (Performance Reports)

1. Module Purpose

The Complaint & Review Management module enables customers to provide feedback after completing their orders while allowing the Customer Service Supervisor to manage complaints, resolve customer issues, and monitor customer satisfaction.

This module helps improve restaurant service quality by collecting customer feedback and ensuring complaints are handled efficiently.

2. Main Responsibilities

This module is responsible for:

- Submitting complaints
- Viewing complaint status
- Updating complaint status
- Resolving complaints
- Submitting reviews
- Editing reviews

- Deleting reviews
 - Viewing customer feedback
-

3. NOT Responsible For

This module **does NOT**:

- ✗ Place customer orders
- ✗ Process restaurant orders
- ✗ Manage deliveries
- ✗ Manage menus
- ✗ Manage branches

Those belong to other modules.

4. Sub Functions

4.1 Submit Complaint

Purpose

Allows customers to report problems related to completed orders.

Examples

- Wrong food item
 - Missing items
 - Poor food quality
 - Late delivery
 - Damaged packaging
-

Frontend

Page

Submit Complaint

Fields

Order Number

Complaint Category

Complaint Description

Optional Image

Submit Button

Backend

Validate the completed order.

Create a new complaint.

Assign initial status.

Database

Table

Complaint

Columns

Complaint_ID

Order_ID

Customer_ID

Category

Description

Image

Status

Created_Date

CRUD

Create

Submit a complaint.

Business Rules

- Complaints can only be submitted for completed orders.
- Customers can submit multiple complaints for different orders.
- Every complaint starts with the status:

Pending

4.2 View Complaint Status

Purpose

Allows customers to monitor the progress of submitted complaints.

Frontend

Page

My Complaints

Displays

Complaint Number

Order Number

Status

Date Submitted

Resolution

CRUD

Read

View complaint status.

Business Rules

Customers can only view their own complaints.

4.3 Update Complaint Status

Purpose

Allows the Customer Service Supervisor to update complaint progress.

Status Flow

Pending

↓

In Progress

↓

Resolved

CRUD

Update

Update complaint status.

Business Rules

Status cannot move backwards.

4.4 Resolve Complaints

Purpose

Allows the Customer Service Supervisor to resolve customer complaints, record the solution, and officially close the complaint.

Frontend

Fields

Resolution Notes

Resolution Date

Button

Resolve Complaint

Backend

Update

Complaint Status

↓

Resolved

Save resolution.

CRUD

Update

Business Rules

- Resolution notes are mandatory.
 - Once resolved, the complaint is closed.
 - Closed complaints cannot be edited.
-

4.5 Submit Review

Purpose

Allows customers to rate and review completed orders.

Frontend

Fields

Rating (1-5 Stars)

Review Comment

Button

Submit Review

Backend

Create review.

Database

Table

Review

Columns

Review_ID

Order_ID

Customer_ID

Rating

Comment

Review_Date

CRUD

Create

Submit review.

Business Rules

- One review per completed order.
 - Reviews can only be submitted for completed orders.
 - Rating is required.
-

4.6 Edit Review

Purpose

Allows customers to modify their own submitted review.

CRUD

Update

Business Rules

- Customers can only edit their own reviews.
 - Reviews cannot be edited after they are deleted.
-

4.7 Delete Review

Purpose

Allows customers to remove their own review.

CRUD

Delete

Business Rules

- Customers can only delete their own reviews.
 - Deleted reviews are no longer displayed to other users.
-

4.8 View Customer Feedback

Purpose

Allows the Customer Service Supervisor to monitor customer satisfaction through ratings and reviews.

Frontend

Dashboard

Displays

Average Rating

Latest Reviews

Lowest Ratings

Highest Ratings

Review Statistics

CRUD

Read

Business Rules

- Customer Service Supervisors can view all reviews.
 - Customers can only view reviews that are publicly displayed (if implemented).
-

5. Database Tables Used

Primary Tables

Complaint

Review

Reference Tables

Order

Customer

Branch

6. APIs

POST /complaints

GET /complaints

GET /complaints/{id}

PATCH /complaints/{id}/status

PATCH /complaints/{id}/resolve

POST /reviews

GET /reviews

PUT /reviews/{id}

DELETE /reviews/{id}

7. Frontend Pages

Submit Complaint

My Complaints

Complaint Details

Complaint Resolution

Submit Review

Edit Review

My Reviews

Customer Feedback Dashboard

8. Integration

Receives Data From

Delivery Management

- Completed delivery status

Uses Data From

Customer Ordering

- Customer information
- Order history

Provides Data To

Branch Management

- Complaint count
- Average customer rating
- Most popular menu items (through reviews and order data)
- Customer satisfaction information for branch performance

9. Testing Checklist

- ☒ Submit complaint
- ☒ Prevent complaint for incomplete orders
- ☒ View complaint status
- ☒ Update complaint status
- ☒ Resolve complaint
- ☒ Submit review
- ☒ Prevent multiple reviews for the same order
- ☒ Edit own review
- ☒ Prevent editing other customers' reviews
- ☒ Delete own review

- ✓ View customer feedback dashboard
-

10. Deliverables

At the end of this module, Member 6 should have completed:

- Complaint Submission
 - Complaint Status Management
 - Complaint Resolution
 - Review Submission
 - Review Editing
 - Review Deletion
 - Customer Feedback Dashboard
 - Frontend pages
 - Backend APIs
 - Database integration
 - Validation
 - Testing
-

Difference Between Member 6 and the Other Members

Module	Responsibility
Member 1	Manages restaurant branches
Member 2	Manages menu items
Member 3	Places customer orders
Member 4	Processes restaurant orders
Member 5	Delivers customer orders
Member 6	Handles customer satisfaction after the order is completed

Simple explanation

- **Member 6 answers:** *"What happens after the customer receives the food?"*

This module closes the business process by collecting customer feedback and resolving service issues.